

Notes: Ru (JF, 2018)

Government Credit, a Double-Edged Sword

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1 Big picture

- **Core message:** Government-directed (policy-bank) credit can *crowd out* private firms in the same industry (by subsidizing SOE expansion), while simultaneously *crowding in* private firms elsewhere through (i) infrastructure productivity gains and (ii) upstream-to-downstream supply-chain spillovers.
1. **Institution + data:** China Development Bank (CDB) provides two big loan categories: industrial (mostly SOEs) and infrastructure.
 2. **Key variation:** city secretary turnover timing creates a predictable “borrow early, then fade” pattern in CDB lending.
 3. **Identification:** predicted (pre-determined) turnover timing $\Rightarrow$ instruments for CDB credit at multiple aggregation levels.
 4. **Main results:**
 - Same-industry: CDB industrial SOE loans expand SOEs and reduce private firm outcomes (crowding out).
 - Downstream: upstream CDB SOE loans raise downstream private outcomes (spillovers).
 - Infrastructure: CDB infrastructure loans raise private outcomes locally (crowding in).

2 Data and key variables (Tables I and AI)

2.1 Datasets and units

- **CDB city-level loans (1998–2010):** city $\times$ year totals, plus industrial vs infrastructure components.
- **CDB province $\times$ industry loans (1998–2013):** province $\times$ industry $\times$ year loan outstanding and issuance (used for industry-level IV and spillovers).
- **Chinese Industrial Census (CIC) firm panel (1998–2009):** manufacturing firms, outcomes:

$$\log(\text{Assets}), \log(\text{Workers}), \log(\text{Debt}), \log(\text{Sales}), \text{ROA}, \log(\text{Sales}/\text{Worker}).$$

- **Politicians:** city secretary tenure and characteristics (age, gender, “Relation” to province governor, promotion).
- **Input–output matrix (2007):** used to map upstream industries into downstream exposure.

2.2 Important constructed indicators (I need to remember these)

- **Focal industry (city-level):** the city’s largest SOE industry (baseline year). This is the bridge from city politics to province×industry loans.
- **SOECity / INFCity:** city types based on whether SOEs dominate the local economy (used to separate industrial vs infrastructure credit in Table IX).
- **CDBCity:** indicator for cities with longer/stronger historical relations with CDB (used to validate that the turnover instrument works through CDB in Table VIII).
- **HighROA and Connected (Table VI):** split downstream private firms by productivity (ROA) and political connections.

3 All empirical models in one place

I list *every* empirical model used in the paper (including the diagnostic models), with the table/figure where it appears.

3.1 Model 0: Cox proportional hazard for turnover timing (Table II)

To check whether predicted turnover timing is related to local economic conditions, the paper estimates Cox hazard models:

$$h_i(t \mid X_{j,t-1}) = h_0(t) \exp(X'_{j,t-1}\beta + Z'_i\gamma), \quad (1)$$

where $X_{j,t-1}$ are lagged city-level conditions (GDP, income, fiscal income, employment, lagged CDB loans), and Z_i are politician traits (age, gender) plus a national-cycle dummy. **Goal:** show that local conditions do not mechanically drive turnover timing (supporting exogeneity of predicted turnover).

3.2 Model 1: Career incentives (promotion) Probit (Appendix Table AII)

$$\Pr(\text{Promotion}_i = 1) = \Phi(\alpha + \delta \log(\Delta \text{LoanIncrease}_i) + \Gamma \cdot \text{Controls}_i). \quad (2)$$

Goal: show that large CDB borrowing increases early in a secretary’s term are rewarded (promotion), which rationalizes why borrowing is front-loaded.

3.3 Model 2: The borrowing cycle at the city level (Eq. (2), Table III, Figure 3)

$$\log(\text{LoanCity}_{j,t}) = \alpha + \sum_{s=2}^6 \beta_s \cdot \text{Year}_{i,j,t}^{(s)} + \theta' \text{Controls}_{j,t-1} + \text{FE} + \varepsilon_{j,t}. \quad (3)$$

Here $\text{Year}_{i,j,t}^{(s)}$ indicates that year t is the s -th year of secretary i 's tenure in city j (Year 1 omitted). The regression is run for total loans, industrial loans, and infrastructure loans, and it is run both for actual tenure timing and predicted tenure timing.

Goal: establish a strong, systematic time pattern: CDB borrowing is highest in Year 1 and then declines.

3.4 Model 3: First stage for province×industry CDB industrial SOE loans (Eq. (3), Appendix Table AIV)

The paper aggregates the city-level turnover indicators to province×industry, exploiting focal-industry assignment.

$$\log(\text{LoanPI}_{k,p,t}) = \alpha + \sum_{s=1}^6 \beta_s \cdot \text{First}_{k,p,t}^{(s)} + \theta' \text{Controls}_{p,t-1} + \text{IndustryFE} + \text{Year} \times \text{ProvinceFE} + \varepsilon_{k,p,t}. \quad (4)$$

Goal: connect turnover timing to the exact credit object used in firm regressions.

3.5 Model 4: Second stage (firm outcomes) for same-industry effects (Eq. (4), Tables IV–V)

$$Y_{l,k,p,t} = \alpha + \beta \cdot \log(\widehat{\text{LoanPI}}_{k,p,t}) + \theta' \text{Controls}_{p,t-1} + \text{FirmFE} + \text{Year} \times \text{ProvinceFE} + u_{l,t}. \quad (5)$$

Goal: estimate causal impacts of CDB industrial SOE credit on firms in the same province×industry.

3.6 Model 5: Supply-chain spillovers (Eqs. (5)–(6), Table VI)

Define for each downstream industry k an upstream focal industry k' using the input–output matrix (conceptually: the most important upstream supplier industry for k). Construct $\text{UpstreamLoan}_{k',p,t}$ as upstream CDB SOE loan outstanding in province p .

First stage (Eq. (5), upstream loan):

$$\begin{aligned} \log(\text{UpstreamLoan}_{k',p,t}) = \alpha + \sum_{s=1}^6 \beta_s \cdot \text{First}_{k',p,t}^{(s)} + \theta' \text{Controls}_{p,t-1} \\ + \text{UpstreamIndustryFE} + \text{Year} \times \text{ProvinceFE} + \varepsilon_{k',p,t}. \end{aligned} \quad (6)$$

Second stage (Eq. (6), downstream firms):

$$Y_{l,k,p,t} = \alpha + \beta \cdot \log(\widehat{\text{UpstreamLoan}}_{k',p,t}) + \theta' \text{Controls}_{p,t-1} + \text{FirmFE} + \text{Year} \times \text{ProvinceFE} + u_{l,t}. \quad (7)$$

3.7 Model 6: Alternative-policy “exclusion” checks (Table VII)

The paper reuses the borrowing-cycle regression (3), but replaces the dependent variable with other potential policy channels: other-bank lending, land development, exports, fiscal income, central government transfers, and tax measures.

Goal: show turnover timing does not predict these policies in the same systematic way, supporting the exclusion restriction.

3.8 Model 7: Stronger turnover channel in CDB-connected cities (Table VIII)

For SOE firm outcomes:

$$Y_{l,j,t} = \alpha + \beta_1 \text{PredictedTurnover}_{j,t} + \beta_2 \text{CDBCity}_j + \beta_3 (\text{PredictedTurnover}_{j,t} \times \text{CDBCity}_j) + \theta' \text{Controls}_{j,t-1} + \text{FirmFE} + \text{YearFE} + u_{l,t}. \quad (8)$$

Goal: show predicted turnover matters *more* where CDB relationships are deeper, reinforcing the mechanism that the political cycle affects outcomes through CDB credit access.

3.9 Model 8: Infrastructure loans vs industrial loans (Table IX, Figure A2)

This is a multi-endogenous-variable IV setup:

$$Y_{l,j,t} = \alpha + \beta_S \log(\widehat{\text{LoanSOECity}}_{j,t}) + \beta_I \log(\widehat{\text{LoanINFCity}}_{j,t}) + \theta' \text{Controls}_{j,t-1} + \text{FirmFE} + \text{YearFE} + u_{l,t}, \quad (9)$$

where the two endogenous regressors are CDB industrial loans and CDB infrastructure loans at the city level. Instruments are interactions between predicted-turnover-year dummies and predetermined city types (SOECity vs INFCity). Figure A2 visualizes the different first-stage sensitivities.

4 Identification logic (what makes the IV credible)

4.1 Relevance

- Table III and Figure 3 show CDB lending is systematically front-loaded in predicted Year 1 of a secretary’s term and declines afterward.
- Appendix Table AIV shows the constructed focal-industry instruments strongly predict province×industry CDB loans (very large first-stage F-statistics).

4.2 Exclusion restriction: what could go wrong, and how the paper responds

The key maintained assumption is:

Predicted turnover timing affects firms only through CDB credit, not through other simultaneous policies.

The paper’s responses:

- **Table II:** predicted turnover timing is not explained by lagged local macro conditions.
- **Table VII:** turnover timing does not systematically move other plausible channels (other-bank loans, land, exports, fiscal/tax).
- **Table VIII:** turnover patterns matter more in historically CDB-connected cities, consistent with a CDB-specific channel.

This is not a proof of exclusion, but it is a coherent set of diagnostics.

Table I
Summary Statistics

Panel A provides summary statistics at the city $\times$ year level for CDB city-level loan data, municipal local economic data, and municipal politician profile data. The data cover 310 cities between 1998 and 2010. Panel B provides summary statistics at the province $\times$ industry $\times$ year level for the CDB province-level loan data, which cover 31 provinces and 95 industries between 1998 and 2013. Panel C provides summary statistics at the firm $\times$ year level for the Chinese Industrial Census data between 1998 and 2009. See Table A1 for detailed variable definitions.

Panel A: City Data					
Variable	N	Mean	SD	Min	Max
Loan_City	3,605	31.97	69.01	0.00	1,268.20
Issuance_City	3,605	11.31	26.76	0.00	428.79
Loan_INF_City	3,605	10.92	30.37	0.00	593.74
Issuance_INF_City	3,605	4.20	12.21	0.00	195.03
Loan_IND_City	3,605	21.34	51.54	0.00	847.77
Issuance_IND_City	3,605	7.13	18.77	0.00	284.22
ActualTurnover	3,605	2.49	1.32	1.00	6.00
PredictedTurnover	3,605	2.62	1.39	1.00	6.00
GDP	3,587	559.46	782.80	0.00	10,604.48
AvgIncome	3,568	10,799.02	8,047.79	0.00	139,574.00
FiscalIncome	3,587	30.43	60.98	0.00	1,138.31
Employment	3,579	1,108.81	18,000.38	0.00	329,858.30
Age	3,325	50.11	4.22	32.00	62.00
Gender	3,605	0.01	0.10	0.00	1.00
Relation	3,158	0.08	0.28	0.00	1.00
Promotion	3,605	0.38	0.48	0.00	1.00

Panel B: Province Data					
Loan_PI	44,733	8.09	43.33	0.00	1,369.09
Issuance_PI	44,733	2.81	16.23	0.00	1,004.12
Loan_Road	496	154.18	163.63	0	744.87
Loan_Rail	496	51.44	102.32	0	1260.99
Loan_Water	496	12.60	23.95	0	203.29
Loan_Tel	496	8.55	41.03	0	419.93

Panel C: Firm Data					
LogAssets	2,949,514	9.72	1.48	0.00	20.16
LogWorkers	2,944,543	4.69	1.17	0.00	12.58
LogDebt	2,930,818	8.98	1.73	0.00	19.32
ROA	2,949,502	0.09	0.20	-0.76	2.44
Log(Sales/W)	2,918,158	5.25	1.24	-8.12	17.38
LogSales	2,931,478	9.95	1.46	0.00	19.24
Tax_Corp	1,520,597	19.41	10.24	0.00	61.54
Tax_VAT	1,356,623	15.09	13.92	0.00	86.91

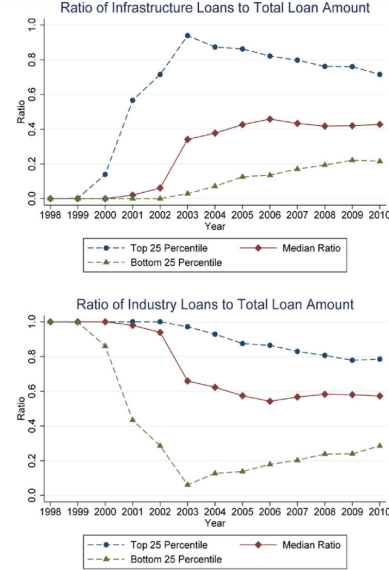


Figure 1. Distribution of CDB infrastructure loans versus industrial SOE loans. This figure plots the distribution of CDB infrastructure loans to total city-level loans (top panel) and the distribution of CDB industrial loans to total city-level loans (bottom panel) across 310 cities. Infrastructure includes transportation (e.g., road, railway, airport, bridge, and tunnel), water supply, energy supply (e.g., gas, electric), telecommunications, and public service (e.g., sewage discharge). City-level loans do not include province-level projects even if part of such projects may be located in the city, such as a highway. The solid lines represent the median ratios across 310 cities and the dashed lines are the top and bottom quartiles of the ratios across 310 cities each year. (Color figure can be viewed at wileyonlinelibrary.com)

5 Tables and figures

I summarize every table and figure in the PDF, emphasizing what to look at, what is estimated, and what conclusion is justified.

5.1 Table I: summary statistics

What it is: descriptive statistics for city-level and province $\times$ industry-level CDB loans, plus firm outcomes and city traits.

What we learn:

- Loan distributions are highly skewed (very large maxima relative to means), so logs are empirically natural.
- Industrial and infrastructure lending are both quantitatively important, supporting the “two-channel” framing.

5.2 Figure 1: loan composition heterogeneity

What it shows: across-city dispersion in infrastructure-loan shares and industrial-loan shares over time (median and interquartile range).

Why it matters: motivates using predetermined city heterogeneity (SOE-heavy vs infrastructure-heavy) to separate the two credit channels in Table IX.

5.3 Table II and Figure 2: turnover timing is not driven by local conditions

Table II (hazard model): lagged GDP, income, fiscal income, employment, and lagged CDB loans do not significantly predict turnover timing; national-cycle years and age do.

Table II
City Secretary Turnover Timing and Local Conditions

This table presents results of Cox proportional hazard regressions following Wooldridge (2002). Data are restricted to 1,106 city secretaries across 310 cities between 1998 and 2010. The origin and failure events are the starts and ends of each city secretary's term. Estimated coefficients are reported in columns (1) to (3). *GDP*, *AvgIncome*, *FiscalIncome*, and *Employment* are city-level GDP, income per capita, fiscal income, and total number of employees, respectively. *Log(Loan_City)* is the logarithm of CDB city-level total loans outstanding. *Age* is the city secretary's age. *Gender* is a dummy for whether the city secretary is female. *NationalCycle* is a dummy for whether the year is a national turnover year (e.g., National Congress Party in 1998, 2003, and 2008). Column (1) covers the full sample. Column (2) focuses on city secretaries who were not promoted at their turnover. Column (3) focuses on city secretaries who were promoted at their turnover. Standard errors are clustered at the city secretary level. T-statistics of the coefficient estimates are reported in parentheses. ***, **, * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

	(1) Coefficient	(2) Coefficient	(3) Coefficient
GDP_{t-1}	-0.142 (0.090)	-0.102 (0.113)	-0.041 (0.128)
$Urban_Income_{t-1}$	0.005 (0.004)	0.004 (0.006)	-0.015 (0.011)
$Fiscal_Income_{t-1}$	1.335 (1.019)	-0.278 (1.801)	1.505 (1.519)
$Labor_{t-1}$	0.410 (0.269)	-5.451 (3.677)	-8.778 (9.390)
$Log(Loan_City_{t-1})$	0.019 (0.020)	0.028 (0.027)	-0.015 (0.032)
<i>NationalCycle</i>	0.343*** (0.067)	0.378*** (0.093)	0.339*** (0.110)
<i>Age</i>	0.029*** (0.008)	0.044*** (0.013)	0.025* (0.014)
<i>Gender</i>	0.055 (0.176)	0.281 (0.364)	0.006 (0.199)
Observations	2,812	1,686	962
χ^2	43.41	47.18	42.45

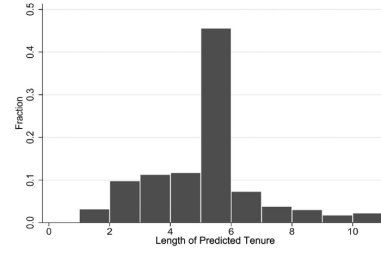


Figure 2. Predicted political turnover of city secretaries in China. This figure plots the distribution of predicted city secretaries' term length in China. The data cover 334 cities and 1,227 city secretaries from 1997 to 2013. Term length is calculated at the city-politician level from the predicted turnover cycles of city secretaries. Approximately 46% of the city secretaries end their terms in the fifth year.

Figure 2: predicted turnover-year distribution concentrates around the institutional cycle (often 5 years). **Interpretation:** the predicted turnover measure is closer to pre-determined timing variation.

5.4 Table III and Figure 3: first-stage borrowing cycle

What is estimated: the borrowing-cycle regression (3) for total, industrial, and infrastructure loans, using both actual and predicted tenure timing.

Key reading rule: Year 1 is the omitted category. A negative β_s means *lower lending relative to Year 1*.

Main fact: coefficients become more negative as s increases, so loans decline over the term. This holds strongly for predicted cycles as well (which is what matters for IV).

Figure 3: visual confirmation of the “zig-zag” pattern in predicted cycles.

5.5 Table IV: OLS correlations (not causal)

Structure:

Table III
City Secretary Turnover and Borrowing from the CDB

This table presents results for regressions of CDB loan amounts on city secretary turnover cycles estimated at the city-year level for 310 cities from 1998 to 2010. Panel A presents results for the OLS regression in equation (2). *Year_2* is a dummy for the second year in a city secretary's term. *Year_3* to *Year_6* are defined similarly. The dummy for year 1 is the missing category. In Panel B, *Turnover* is the number of years a secretary has served the city. Columns (1) to (3) give the effects of the actual turnover cycles on city-level CDB total loans outstanding, infrastructure loans, and industrial SOE loans, respectively. Columns (4) to (6) give the corresponding effects of the predicted turnover cycles. $\text{LogLoan} = \log(\text{Loan.City})$, which is the logarithm of city-level CDB total loans outstanding. $\text{LogLoanINF} = \log(\text{Loan.INF.City})$, which is the logarithm of infrastructure loans. $\text{LogLoanIND} = \log(\text{Loan.IND.City})$, which is the logarithm of industrial SOE loans. Control variables include city-level GDP, income per capita, fiscal income, and total number of employees in year $t - 1$. All columns also control for year, city, and politician fixed effects. Standard errors are clustered at the city level. T-statistics of the coefficient estimates are reported in parentheses. ***, **, * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

Dependent Variable	Actual Turnover			Predicted Turnover		
	(1) LogLoan	(2) LogLoanINF	(3) LogLoanIND	(4) LogLoan	(5) LogLoanINF	(6) LogLoanIND
Panel A: Turnover Dummies						
Year_2	-0.386*** (0.031)	-0.109*** (0.028)	-0.330*** (0.044)	-0.182*** (0.030)	-0.117*** (0.032)	-0.149*** (0.040)
Year_3	-0.749*** (0.042)	-0.240*** (0.035)	-0.703*** (0.064)	-0.303*** (0.034)	-0.176*** (0.039)	-0.336*** (0.051)
Year_4	-1.071*** (0.063)	-0.351*** (0.041)	-1.001*** (0.086)	-0.473*** (0.037)	-0.231*** (0.044)	-0.495*** (0.052)
Year_5	-1.429*** (0.108)	-0.467*** (0.081)	-1.341*** (0.151)	-0.635*** (0.071)	-0.343*** (0.067)	-0.645*** (0.095)
Year_6	-1.900*** (0.144)	-0.691*** (0.110)	-1.668*** (0.204)	-0.729*** (0.116)	-0.451*** (0.096)	-0.680*** (0.159)
Control&FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	3,127	2,449	2,727	3,161	2,429	2,759
R ²	0.854	0.884	0.756	0.845	0.860	0.764
Panel B: Turnover Years						
Turnover	-0.364*** (0.020)	-0.122*** (0.013)	-0.338*** (0.028)	-0.154*** (0.012)	-0.080*** (0.012)	-0.160*** (0.015)
Control&FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	3,127	2,449	2,727	3,161	2,429	2,759
R ²	0.853	0.884	0.756	0.845	0.859	0.763

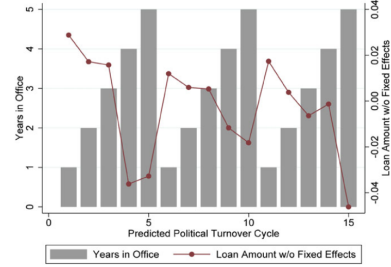


Figure 3. Local government borrowing pattern. This figure plots the logarithm of CDB total city loan amounts over city secretaries' predicted turnover cycles. The right vertical axis is the logarithm of CDB total city loans after removing the year, city, and politician fixed effects from the regressions. The horizontal axis is the predicted turnover cycle of city secretaries. There are three national five-year turnover cycles between 1998 and 2012: 1998 to 2002, 2003 to 2007, and 2008 to 2012. The left vertical axis is the number of years that a city secretary serves the city (predicted), which is from year 1 to year 5. For example, for the first cycle (1 to 5 on the horizontal axis) from 1998 to 2002, I cluster cities by "Years in Office" (1 to 5) from 1998 to 2002 and plot the average CDB city loan amounts for each "Years in Office" bin. I repeat this procedure for the second cycle (6 to 10 on the horizontal axis) from year 2003 to 2007. The third cycle (11 to 15 on the horizontal axis) has only three years from 2008 to 2010, as the CDB city-level loan data are between 1998 and 2010. (Color figure can be viewed at wileyonlinelibrary.com)

- Panel A: SOE outcomes vs industrial SOE loans (positive correlations).
- Panel B: private outcomes in same industry vs industrial SOE loans (negative correlations).
- Panel C: private outcomes vs infrastructure loans (positive correlations).

Interpretation: correlations line up with hypotheses, but endogeneity concerns remain (hence 2SLS in Table V and Table IX).

5.6 Table V: 2SLS same-industry effects

Private firms (Panel A): higher CDB industrial SOE loans reduce private firm size and performance in the same industry. Example (assets): $\beta \approx -0.029$. A doubling of CDB SOE loans implies:

$$\Delta \log(\text{Assets}) \approx -0.029 \cdot \log 2 \approx -0.020 \Rightarrow \text{about a 2\% decline.}$$

SOEs (Panel B): higher CDB industrial SOE loans expand SOEs strongly (large positive coefficients on assets, workers, debt, sales). **Economic meaning:** same-industry crowding out is a relative-shares story: subsidized SOE expansion squeezes private firms.

5.7 Table VI: 2SLS downstream spillovers

Panel A: average downstream effect. Upstream CDB SOE loans increase downstream private assets and sales and raise profitability and sales per worker.

Table IV
Effects of CDB Loans on Firms (OLS)

This table presents results for regressions of firm activities on CDB loan amounts estimated at the firm-year level from 1998 to 2009. In Panels A and B, *Loan_PI* is CDB industrial SOE loans outstanding for each of the 31 provinces and 41 manufacturing industries per year. In Panel A, the sample is restricted to SOEs in the CIC data. In Panel B, the sample is restricted to private firms. *Control₋₁* includes province-level GDP, income per capita, fiscal income, and total number of employees in year $t-1$. In Panel C, *Loan_INF_City* is CDB infrastructure loans outstanding at the city-year level. The sample is restricted to private firms in the CIC data from 1998 to 2009 across 310 cities. *Control₋₁* includes city-level GDP, income per capita, fiscal income, and total number of employees in year $t-1$. All regressions also control for firm fixed effects and year-province fixed effects. Standard errors are clustered at the firm level. T-statistics of the coefficient estimates are reported in parentheses. ***, **, * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

Dependent Variable	(1) LogAssets	(2) LogWorkers	(3) LogDebt	(4) LogSales	(5) ROA	(6) Log(Sales/W)
Panel A: SOEs						
Log(Loan_PI)	0.011*** (0.002)	0.011*** (0.002)	0.023*** (0.003)	0.005* (0.003)	-0.001*** (0.000)	-0.003 (0.003)
Control ₋₁	Yes	Yes	Yes	Yes	Yes	Yes
Fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	231,682	232,003	229,696	231,682	225,214	227,342
R ²	0.119	0.046	0.076	0.174	0.021	0.206
Panel B: Private Firms						
Log(Loan_PI)	-0.006*** (0.001)	-0.004*** (0.001)	-0.000 (0.001)	-0.017*** (0.001)	-0.003*** (0.000)	-0.013*** (0.001)
Control ₋₁	Yes	Yes	Yes	Yes	Yes	Yes
Fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	756,826	757,033	752,871	755,947	756,820	755,687
R ²	0.243	0.014	0.101	0.282	0.030	0.237
Panel C: Infrastructure						
Log(Loan_INF_City)	0.013*** (0.003)	-0.004* (0.002)	0.016*** (0.003)	0.033*** (0.003)	0.002*** (0.001)	0.036*** (0.002)
Control ₋₁	Yes	Yes	Yes	Yes	Yes	Yes
Fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,022,595	1,022,564	1,016,623	1,022,586	1,021,059	1,021,245
R ²	0.210	0.023	0.085	0.278	0.051	0.242

Table V
Effects of CDB SOE Loans on Firms in Same Industry (2SLS)

This table presents two-stage least squares regression results using *First* to *Sixth* as instrumental variables for the logarithm of CDB province-level loans outstanding in 41 manufacturing industries and 27 provinces (excluding Beijing, Shanghai, Tianjing, and Chongqing). In Panel A, the sample is restricted to private firms in the CIC data from 1998 to 2009. In Panel B, the sample is restricted to SOEs in the CIC data. *Loan_PI* is CDB loans outstanding in the fiscal industry, which is at the province-industry-year level. *Control₋₁* includes province-level GDP, income per capita, fiscal income, and total number of employees in year $t-1$. I also control for CDB infrastructure loans outstanding. All columns control for firm fixed effects and year-province fixed effects. Standard errors are clustered at the firm level. Cragg-Donald Wald F-statistics for weak identification tests are reported. T-statistics of the coefficient estimates are reported in parentheses. ***, **, * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

Dependent Variable	(1) LogAssets	(2) LogWorkers	(3) LogDebt	(4) LogSales	(5) ROA	(6) Log(Sales/W)
Panel A: Private Firms						
Log(Loan_PI)	-0.029*** (0.009)	-0.025*** (0.008)	-0.007 (0.013)	-0.060*** (0.011)	-0.005* (0.003)	-0.038*** (0.010)
Control ₋₁	Yes	Yes	Yes	Yes	Yes	Yes
Fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Wald F-stat	719.989	719.271	715.834	719.140	718.786	717.844
	643.2	642.5	648.4	643.4	642.3	642.4
Panel B: SOEs						
Log(Loan_PI)	0.355*** (0.020)	0.209*** (0.018)	0.314*** (0.026)	0.198*** (0.023)	-0.010** (0.004)	0.063*** (0.021)
Control ₋₁	Yes	Yes	Yes	Yes	Yes	Yes
Fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	230,645	231,657	228,465	225,879	230,628	224,067
Wald F-stat	179.8	185.1	179.5	179.1	179.9	176.6

Table VI
Effects of Upstream CDB SOE Loans on Firms in Downstream Industries (2SLS)

This table presents two-stage least squares regression results using *First* to *Sixth* as instrumental variables for the logarithm of CDB province-level loans outstanding in 25 manufacturing industries (colapsed from 41 CDB manufacturing industries) and 27 provinces (including Beijing, Shanghai, Tianjing, and Chongqing). The sample is restricted to private firms included in the CIC data from 1998 to 2009. *Upstream_Loan* is CDB industrial SOE loans outstanding in a firm's upstream focal industry, which is at the province-industry-year level. In Panel B, *HighROA* is a dummy for whether the firm's ROA is above the median across all private firms in the previous year. In Panel C, *Connected* is a dummy for whether the private firm's political hierarchy is above the city level (i.e., province level or national level); it captures firm political connections. I follow Woolfridge (2002) to include the interaction term in 2818. *Control₋₁* includes province-level GDP, income per capita, fiscal income, and total number of employees in year $t-1$. I also control for CDB infrastructure loans outstanding. All columns also control for firm fixed effects and year-province fixed effects. Standard errors are clustered at the firm level. Cragg-Donald Wald F-statistics for weak identification tests are reported. T-statistics of the coefficient estimates are reported in parentheses. ***, **, * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

Dependent Variable	(1) LogAssets	(2) LogWorkers	(3) LogDebt	(4) LogSales	(5) ROA	(6) Log(Sales/W)
Panel A: Upstream Loans						
Log(Upstream_Loan)	0.049*** (0.009)	-0.001 (0.008)	0.008 (0.009)	0.037*** (0.007)	0.007*** (0.002)	0.037*** (0.007)
Control ₋₁	Yes	Yes	Yes	Yes	Yes	Yes
Fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	804,718	804,853	799,002	803,712	804,709	803,296
Wald F-stat	1,601	1,604	1,593	1,600	1,601	1,602
Panel B: High ROA Firms						
Log(Upstream_Loan)	-0.006 (0.006)	-0.051*** (0.005)	-0.001 (0.009)	-0.004 (0.007)	0.006*** (0.002)	-0.008 (0.007)
HighROA	-0.130*** (0.003)	-0.072*** (0.003)	-0.161*** (0.005)	-0.174*** (0.004)	-0.062* (0.001)	-0.103*** (0.004)
Log(Upstream_Loan)	0.131*** (0.003)	0.085*** (0.004)	0.106*** (0.004)	0.158*** (0.001)	0.002*** (0.001)	0.073*** (0.003)
Control ₋₁	Yes	Yes	Yes	Yes	Yes	Yes
Fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	804,718	804,853	799,002	803,712	804,709	803,296
Wald F-stat	3,388	3,397	3,366	3,388	3,388	3,397
Panel C: Connected Firms						
Log(Upstream_Loan)	0.027*** (0.006)	-0.001 (0.005)	0.007 (0.009)	0.037*** (0.007)	0.007*** (0.002)	0.038*** (0.007)
Log(Upstream_Loan)	0.058*** (0.019)	0.062*** (0.014)	0.123*** (0.023)	0.017 (0.019)	-0.021*** (0.003)	-0.045*** (0.014)
Connected	Yes	Yes	Yes	Yes	Yes	Yes
Control ₋₁	Yes	Yes	Yes	Yes	Yes	Yes
Fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	285,544	283,880	282,245	285,544	275,484	279,226
Wald F-stat	620.1	607.8	602.5	617.9	577.8	590.6

Panel B: heterogeneity by productivity (HighROA). If the regression includes $\log(\text{UpstreamLoan})$ and $\log(\text{UpstreamLoan}) \times \text{HighROA}$, then

$$\text{Effect for low-ROA firms} = \beta_{\text{loan}}, \quad \text{Effect for high-ROA firms} = \beta_{\text{loan}} + \beta_{\text{loan} \times \text{HighROA}}.$$

The table shows the positive scale effects are concentrated among high-ROA firms, consistent with spillovers enabling efficient firms to expand.

Panel C: heterogeneity by political connections (Connected). Connected firms appear to expand on some margins but the efficiency gains (ROA, sales per worker) weaken or reverse, consistent with political connections affecting how spillovers translate into measured productivity.

5.8 Table VII: other channels are not systematically tied to turnover timing

What it does: uses the borrowing-cycle specification but swaps in alternative dependent variables (other-bank loans, land, exports, fiscal income, transfers, taxes).

What I conclude: the joint F-tests for the Year dummies are usually insignificant, so turnover timing is not mechanically driving those policies in the same systematic way. This supports (but does not prove) the exclusion restriction.

5.9 Table VIII: CDBCity interaction test

What it shows: the negative relationship between “years in predicted cycle” and SOE outcomes is stronger in cities with longer-standing CDB relations (negative and significant interaction terms).

Why it matters: this is consistent with the instrument operating through a CDB-specific lending channel, rather than a generic political-cycle channel.

Table VII

Other Channels through Which Politicians Affect the Local Economy

This table repeats the regressions in Table III (i.e., equation (2)) but regress other channels through which city secretaries can influence the local economy on predicted turnover cycles. The data cover 310 cities between 1998 and 2010. In column (1), the dependent variable is the logarithm of the city's loans outstanding from other banks. The data come from the CBRC from 2007 to 2009. In column (2), *Land* is the total developed land in each city every year. In column (3), *Export* is the total export amount in each city-year. In column (4), *Fiscal_Income* is the total fiscal income in each city-year. In column (5), *Transfer* is the amount of fiscal transfers in each city-year. In column (6), *Tax_Corp* is the average manufacturing firm's effective corporate tax rate in each city-year. In column (7), *Tax_VAT* is the average manufacturing firm's effective value-added tax rate in each city-year. *Control_{t-1}* includes city level GDP, income per capita, fiscal income, and total number of employees in year $t - 1$. All columns also control for year, city, and politician personal fixed effects. Standard errors are clustered at the city level. The table also reports F -statistics and p -values for the F -tests on the turnover dummies *Year_2* to *Year_6*. T-statistics of the coefficient estimates are reported in parentheses. ***, **, * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

Dependent Variable	(1) Log(OtherLoans)	(2) Land	(3) Export	(4) Fiscal_Income	(5) Transfer	(6) Tax_Corp	(7) Tax_VAT
Year_2	-0.044 (0.092)	-0.009 (0.011)	0.802 (1.530)	-0.387 (0.608)	0.355 (0.582)	-0.038 (0.166)	0.111 (0.348)
Year_3	-0.028 (0.112)	0.002 (0.021)	1.414 (2.407)	-0.855 (0.838)	0.242 (0.683)	0.260 (0.198)	-0.051 (0.315)
Year_4	0.095 (0.158)	-0.006 (0.016)	0.266 (0.881)	-0.212 (0.588)	-0.527 (0.541)	0.166 (0.210)	-0.261 (0.288)
Year_5	0.253 (0.237)	-0.004 (0.015)	1.138 (1.664)	-0.455 (0.724)	-0.418 (0.373)	-0.221 (0.305)	0.058 (0.331)
Year_6	0.031 (0.305)	0.007 (0.021)	0.770 (2.476)	-1.588* (0.888)	-1.136 (0.996)	-0.613 (0.476)	0.066 (0.477)
Control _{t-1}	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	861	3,319	3,494	3,499	3,306	3,181	1,819
R^2	0.453	0.233	0.409	0.968	0.915	0.513	0.692
F -statistics	0.93	0.64	0.25	1.22	0.42	1.26	0.53
p -values	0.46	0.67	0.94	0.30	0.84	0.28	0.75

Table VIII

Effects of Turnover Timing across Cities with Different Relationships to the CDB

This table presents results for regressions of firm activities on predicted political turnover cycles across cities with different relationships with the CDB. The sample is restricted to SOEs included in the CIC data from 1998 to 2009. *PredictedTurnover* is the number of years that a secretary serves the city in the predicted turnover cycles. *CDBCity* is a dummy for whether the city borrowed from the CDB in 1998. This predetermined dummy captures the relationship between the CDB and cities. All columns control for year $\times$ province fixed effects, firm fixed effects, and politician personal fixed effects. *Control_{t-1}* includes city-level GDP, income per capita, fiscal income, and total number of employees in year $t - 1$. Standard errors are clustered at the firm level. T-statistics of the coefficient estimates are reported in parentheses. ***, **, * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

Dependent Variable	(1) LogAssets	(2) LogWorkers	(3) LogDebt	(4) LogSales	(5) ROA	(6) Log(Sales/W)
PredictedTurnover	0.004 (0.008)	0.002 (0.007)	0.012 (0.010)	-0.004 (0.010)	0.000 (0.002)	-0.001 (0.010)
CDBCity	0.091 (0.063)	0.076 (0.060)	0.137 (0.084)	0.089 (0.074)	-0.037** (0.016)	0.014 (0.071)
PredictedTurnover $\times$	-0.020*** (0.003)	-0.014*** (0.003)	-0.024*** (0.004)	-0.015*** (0.004)	0.004*** (0.001)	-0.002 (0.004)
Control _{t-1}	Yes	Yes	Yes	Yes	Yes	Yes
Fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	396,910	395,802	392,876	387,132	396,911	382,790
R^2	0.112	0.074	0.067	0.151	0.041	0.181

5.10 Table IX and Figure A2: infrastructure vs industrial loans

Table IX: in a horse race, industrial SOE loans have negative effects on private outcomes, while infrastructure loans have positive effects on private outcomes (assets, employment, debt, sales, ROA, sales per worker).

Table IX

Opposing Effects of CDB City Infrastructure Loans versus Industrial Loans on Private Firms (2SLS)

This table presents two-stage least squares results using interactions between predicted *Year_1* to *Year_6* and the dummy *SOECity* as instrumental variables for both CDB infrastructure annual loan amounts and CDB industrial SOE annual loan amounts at the city level. In the first stage, I regress the logarithm of CDB city-level industrial SOE loan amounts on *SOEYear_1* to *SOEYear_5* and *INFYear_1* to *INFYear_5*, where *SOEYear_1* = *SOECity* $\times$ *Year_1*, *INFYear_1* = *INFCity* $\times$ *Year_1*, etc. I control for the same variables and fixed effects as in the second-stage regressions. I perform the same first-stage regressions on CDB city-level infrastructure loan amounts. *Loan_IND.City* is aggregate CDB loans outstanding for infrastructure at the city $\times$ year level. *Loan_INF.City* is aggregate CDB loans outstanding for industry firms at the city $\times$ year level. The sample is restricted to private firms included in the CIC data from 1998 to 2009 across 310 cities. *Control_{t-1}* includes city-level GDP, income per capita, fiscal income, and total number of employees in year $t - 1$. All columns control for the year $\times$ province fixed effects, firm fixed effects, and politician personal fixed effects. Standard errors are clustered at the firm level. Cragg-Donald Wald F -statistics for weak identification tests are reported. T-statistics of the coefficient estimates are reported in parentheses. ***, **, * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

Dependent Variable	(1) LogAssets	(2) LogWorkers	(3) LogDebt	(4) LogSales	(5) ROA	(6) Log(Sales/W)
Log(Loan_IND_City)	-0.020* (0.011)	-0.019** (0.009)	0.037** (0.015)	-0.039*** (0.012)	-0.024*** (0.003)	-0.021** (0.010)
Log(Loan_INF_City)	0.135*** (0.014)	0.028** (0.012)	0.095*** (0.021)	0.315*** (0.016)	0.037*** (0.004)	0.289*** (0.015)
Control _{t-1}	Yes	Yes	Yes	Yes	Yes	Yes
Fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	859,086	859,125	854,012	858,221	859,080	857,846
Wald F -stat	1,160	1,148	1,156	1,158	1,159	1,147

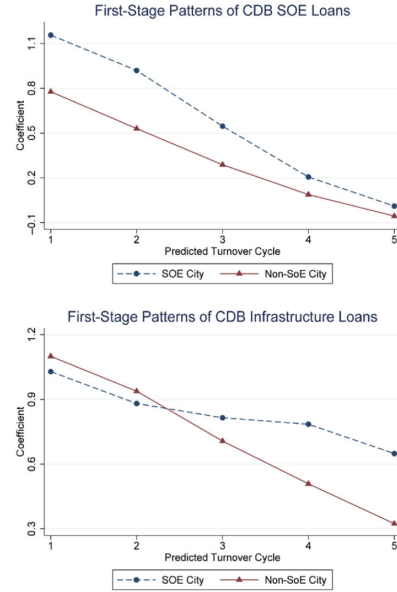


Figure A2. First-stage patterns of CDB SOE loans versus infrastructure loans. This figure plots regression coefficients for the first-stage regressions in Table IX. The top panel shows the regression coefficients for the instruments *SOECity* $\times$ *Year_1* to *SOECity* $\times$ *Year_5* and $(1 - SOECity) \times Year_1$ to $(1 - SOECity) \times Year_5$ on the CDB SOE loan amount. The bottom panel shows the regression coefficients for the instruments *SOECity* $\times$ *Year_1* to *SOECity* $\times$ *Year_5* and $(1 - SOECity) \times Year_1$ to $(1 - SOECity) \times Year_5$ on the CDB infrastructure loan amount. (Color figure can be viewed at wileyonlinelibrary.com)

Interpretation:

- Industrial credit to SOEs can be distortionary for private firms in the same local economy.

- Infrastructure credit can raise the private sector's effective productivity locally.

Figure A2: validates the instrument design by showing the first-stage sensitivity differs across city types (SOE vs infrastructure-heavy).

5.11 Figure 4: time variation in industrial policy priorities

What it shows: the identity of top borrowing industries shifts over time. This helps rationalize why aggregate or average effects of government credit can change with policy-bank priorities.

5.12 Appendix Table AI: variable definitions

Use: a dictionary of variables, units, and sample coverage (especially for taxes and other firm-level variables with missing years).

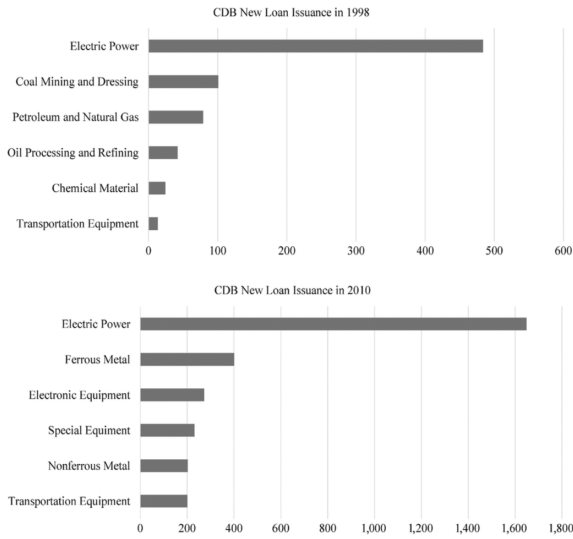


Figure 4. Shifts of CDB industrial loans over time. This figure plots the six industries that receive the largest CDB credit inflows (i.e., loan issuance). Data are restricted to CDB province-level industrial loans in 41 manufacturing industries across 31 provinces in China. The top panel shows CDB loan issuance amounts to the top six industries in 1998. The amount for each industry is the sum of all CDB loan issuance amounts across 31 provinces in China. The bottom panel shows CDB loan issuance amounts to the top six industries in 2010. The units are billions RMB.

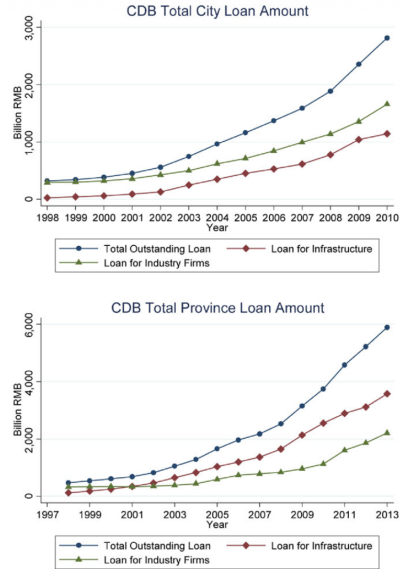


Figure A1. Time trend of CDB outstanding loan amounts. This figure plots the CDB outstanding loan amounts. The loans are separated into infrastructure loans and loans for industrial firms. Infrastructure includes transportation (e.g., road, railway, airport, bridge, and tunnel), water supply, energy supply (e.g., gas, electric), telecommunications, and public service (e.g., sewage discharge). The top panel corresponds to city-level loans and the bottom panel to province-level loans. City-level loans do not include province-level projects even if part of the project is located in the city, such as a highway. Units are in billions RMB. (Color figure can be viewed at wileyonlinelibrary.com)

5.13 Appendix Table AII: promotion results

Conclusion: larger CDB loan increases early in the term are positively associated with promotion probabilities, supporting the career-concerns mechanism behind front-loaded borrowing.

5.14 Appendix Table AIII: robustness excluding national-cycle cities

Conclusion: the borrowing-cycle pattern remains when excluding cities aligned with national turnover years, suggesting the first stage is not purely a national-cycle artifact.

Table AII
City Secretary Promotion and CDB Loans

This table presents the results from Probit regressions of a city secretary's promotion on CDB loan amounts. City secretaries' political hierarchies vary across cities. Secretaries from Beijing, Shanghai, Tianjin, and Chongqing are at the ministerial level. Secretaries from 15 cities in China are at the vice-ministerial level: Dalian, Qingdao, Ningbo, Xiamen, Shenzhen, Haerbing, Changchun, Shenyang, Jinan, Nanjing, Hangzhou, Guangzhou, Wuhan, Chengdu, and Xian. All others are at the departmental level. I exclude the ministerial-level cities because they are at the same level as provinces. *Promotion* is a dummy that indicates whether a city secretary is promoted based on their political hierarchy. $\text{Log}(\text{Loan Increase in 1st year})$ is the logarithm of the increase in CDB loans outstanding in the first year of the city secretary's tenure. $\text{Log}(\text{Loan Increase in 1st} - 2\text{nd year})$ is the logarithm of the increase in CDB loans outstanding in the first two years of the city secretary's tenure. *Age* is the age of the city secretary at the end of their term. *Relation* is a dummy for whether the city secretary is born in the same city as the province governor. *Gender* is a dummy for whether the city secretary is female. Standard errors are clustered at the city level. Z-statistics of the coefficient estimates are reported in parentheses. ***, **, * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

Dependent Variable	(1) Promotion	(2) Promotion	(3) Promotion	(4) Promotion	(5) Promotion
Log(Loan Increase in 1st year)	0.102*** (0.035)				
Log(Loan Increase in 1st–2nd year)		0.087** (0.034)			
Log(Loan Increase in 1st–3rd year)			0.055* (0.033)		
Log(Loan Increase in 1st–4th year)				0.062* (0.032)	
Log(Loan Increase in 1st–5th year)					0.057* (0.033)
Relation	0.430* (0.237)	0.457** (0.218)	0.431** (0.211)	0.394* (0.207)	0.426** (0.205)
Age	−0.084*** (0.013)	−0.078*** (0.013)	−0.074*** (0.012)	−0.074*** (0.012)	−0.074*** (0.012)
Gender	0.617 (0.388)	0.663* (0.388)	0.477 (0.364)	0.410 (0.381)	0.407 (0.381)
Observations	691	758	792	810	812
χ^2	54.58	48.75	42.96	43.85	43.70

Table AIII
City Secretary Turnover and Borrowing from the CDB
(Off-National-Cycle Cities)

This table repeats the regressions in Table III, but excludes city secretaries whose turnover cycles are in the same year as the national turnover cycle (i.e., 1998, 2003, and 2008), that is, the sample limits to cities with turnover cycles that differ from national cycle. *Year 2* is a dummy for the second year in a city secretary's term. *Year 3* to *Year 6* are defined in the same way. The dummy for year 1 is the missing category. Columns (1) to (3) are for the effect of actual turnover cycle on the total CDB city loans outstanding, infrastructure loans outstanding, and industrial SOE loans outstanding, respectively. Columns (4) to (6) are for the effect of predicted turnover cycle. $\text{LogLoan} = \text{log}(\text{Loan.City})$, which is the logarithm of CDB total loans outstanding at the city-level. $\text{LogLoanINF} = \text{log}(\text{Loan.INF.City})$, which is the logarithm of infrastructure loans outstanding at the city-level. $\text{LogLoanIND} = \text{log}(\text{Loan.IND.City})$, which is the logarithm of industrial SOE loans outstanding at the city-level. Control variables include city-level GDP, income per capita, fiscal income, and total number of employees in year $t - 1$. All columns are controlled by year, city, and politician fixed effects. Standard errors are clustered at the city level. T-statistics of the coefficient estimates are reported in parentheses. ***, **, * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

Dependent Variable	Actual Turnover			Predicted Turnover		
	(1) LogLoan	(2) LogLoanINF	(3) LogLoanIND	(4) LogLoan	(5) LogLoanINF	(6) LogLoanIND
Year 2	−0.086* (0.047)	−0.119*** (0.042)	−0.043 (0.063)	−0.155*** (0.036)	−0.229*** (0.048)	−0.026 (0.049)
Year 3	−0.137*** (0.063)	−0.228*** (0.045)	−0.179** (0.083)	−0.279*** (0.050)	−0.379*** (0.082)	−0.182** (0.089)
Year 4	−0.181** (0.091)	−0.358*** (0.055)	−0.207* (0.119)	−0.468*** (0.050)	−0.512*** (0.111)	−0.370*** (0.094)
Year 5	−0.314** (0.154)	−0.487*** (0.095)	−0.415* (0.217)	−0.665*** (0.078)	−0.729*** (0.130)	−0.466*** (0.102)
Year 6	−0.579*** (0.195)	−0.728*** (0.133)	−0.611** (0.278)	−0.793*** (0.132)	−0.966*** (0.191)	−0.459*** (0.164)
Control _{t−1}	Yes	Yes	Yes	Yes	Yes	Yes
Fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,686	1,387	1,495	2,108	1,705	1,835
R ²	0.810	0.856	0.707	0.827	0.833	0.711

5.15 Appendix Table AIV: first-stage estimates

Conclusion: the focal-industry turnover instruments strongly predict province×industry CDB loans (the key first-stage needed for Table V).

5.16 Appendix Figure A1: CDB loan time trend

Interpretation: CDB lending expands over time and its composition evolves. This is context for why one should not interpret government credit effects as time-invariant.

6 Short summary

1. **Instrument:** predicted city secretary tenure timing $\Rightarrow$ front-loaded CDB borrowing $\Rightarrow$ predicted variation in CDB loans.
2. **Same-industry result:** CDB industrial SOE loans expand SOEs and crowd out private firms in that industry (Table V).
3. **Spillover result:** upstream CDB loans crowd in downstream private firms (Table VI).
4. **Infrastructure result:** CDB infrastructure loans crowd in private firms locally (Table IX).

Table AIV
CDB Province-Industry-Level Loan and City Secretary Turnover
(First Stage)

This table presents results of regressions of CDB province-industry loan amounts on the *First* to *Sixth* dummies at the province $\times$ industry $\times$ year level. The sample is restricted to CDB province-level industrial loans outstanding in 31 provinces and 41 manufacturing industries from 1998 to 2009. *Log(Loan_PI)* is the logarithm of CDB annual province-industry loans outstanding. *First* is a dummy for whether the city secretary is in the predicted first year of her term and the firm is in the city's largest SOE industry (i.e., focal industry). *Second* is a dummy for whether the city secretary is in the predicted second year of the term and the firm is in the city's focal industry. The dummies *Third* to *Sixth* are defined similarly. *FirstSecond* is a dummy for *First* or *Second* equals one. *FirstThird* is a dummy for *First*, *Second*, or *Third* equals one. *Control_{t-1}* includes city-level GDP, income per capita, fiscal income, and total number of employees in year $t - 1$. All columns are controlled by industry fixed effects and year $\times$ province fixed effects. Standard errors are clustered at the province level. T-statistics of the coefficient estimates are reported in parentheses. ***, **, * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

Dependent Variable	(1) Log(Loan_PI)	(2) Log(Loan_PI)	(3) Log(Loan_PI)
First	0.341*** (0.118)		
Second	0.285** (0.105)		
Third	0.274*** (0.090)		
Fourth	0.260** (0.115)		
Fifth	0.211** (0.100)		
Sixth	0.044 (0.131)		
FirstSecond		0.413*** (0.134)	
FirstThird			0.386*** (0.119)
Control _{t-1}	Yes	Yes	Yes
Fixed effects	Yes	Yes	Yes
Observations	4,445	4,445	4,445
R ²	0.564	0.561	0.561

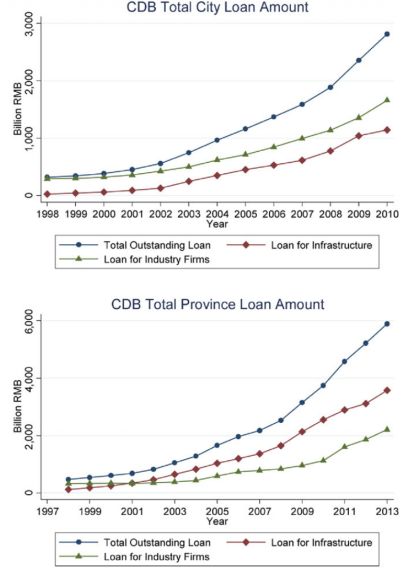


Figure A1. Time trend of CDB outstanding loan amounts. This figure plots the CDB outstanding loan amounts. The loans are separated into infrastructure loans and loans for industrial firms. Infrastructure includes transportation (e.g., road, railway, airport, bridge, and tunnel), water supply, energy supply (e.g., gas, electric), telecommunications, and public service (e.g., sewage discharge). The top panel corresponds to city-level loans and the bottom panel to province-level loans. City-level loans do not include province-level projects even if part of the project is located in the city, such as a highway. Units are in billions RMB. (Color figure can be viewed at wileyonlinelibrary.com)

7 Conclusion

7.1 Summary

This paper studies a central question in development finance and political economy: *does government-directed credit crowd out or crowd in the private sector?* Using proprietary loan data from the China Development Bank (CDB) and tracing effects along production networks, the paper shows that government credit is a **double-edged sword**.

The core conclusion is that the sign of the effect depends on (i) **who receives the credit** and (ii) **where a firm sits in the supply chain**:

- **Industrial loans to SOEs** expand SOEs but **crowd out private firms in the same industry** (direct competition + resource reallocation), while **crowding in private firms in downstream industries** (via supply-chain spillovers).
- **Infrastructure loans to local governments** have broad **crowding-in effects** on private firms (productivity/market-access spillovers).

Because these effects operate simultaneously, the paper argues that evaluating government credit requires looking beyond the direct recipients and accounting for **network spillovers** and **credit type**.

7.2 Key empirical conclusions (with quantitative anchors)

1. **Direct crowding-out in the same industry.** A doubling (100% increase) of CDB industrial SOE loan amounts to a local “focal” industry leads to decreases for private firms in the *same*

province–industry of about:

$$\Delta \text{Assets} \approx -2.0\%, \quad \Delta \text{Employment} \approx -1.7\%, \quad \Delta \text{Sales} \approx -4.1\%.$$

The paper also finds deterioration in profitability/efficiency measures for same-industry private firms (e.g., ROA and sales per worker), consistent with distorted competition and/or misallocation.

2. **Indirect crowding-in downstream (supply-chain propagation).** Using China’s input–output matrix to define downstream industries, a doubling of CDB credit to the focal (upstream) industry increases downstream private-firm outcomes (within the same province) by roughly:

$$\Delta \text{Assets} \approx +3.4\%, \quad \Delta \text{Sales} \approx +2.6\%, \quad \Delta \text{Sales per worker} \approx +2.6\%.$$

The paper further documents that **more efficient downstream private firms benefit more**, consistent with spillovers that are amplified when firms can scale up quickly.

3. **Infrastructure credit crowds in the private sector.** In contrast to industrial SOE lending, infrastructure lending has broad positive effects. A back-of-the-envelope interpretation in the paper implies that, on average,

$$\text{\$1 increase in CDB infrastructure loans} \Rightarrow \text{about \$0.47 increase in private firms' assets.}$$

In productivity terms, infrastructure loans are associated with large improvements in sales per worker for private firms (the paper summarizes an increase on the order of $\sim 18\%$ based on estimated coefficients).

4. **Net effects are time-varying because CDB lending shifts across industries.** The paper shows that the same type of CDB industrial lending can look “pro-private” in earlier years and “anti-private” later, consistent with the CDB shifting its industrial-loan focus across upstream/basic industries versus other industries over time. This provides a concrete explanation for why aggregate studies can find mixed crowding-in vs crowding-out results.
5. **Political incentives are central to credit allocation (identification + interpretation).** The paper exploits predetermined political turnover cycles of municipal leaders as an instrument for CDB credit flows, consistent with a “zig-zag” borrowing pattern around leader tenure. This supports the idea that government credit is not purely technocratic—it is shaped by political incentives and thus can generate systematic reallocations.

7.3 Economic intuition: why the results fit together

A coherent mechanism is a **direct-competition channel** plus a **network-spillover channel**:

- **Industrial SOE loans: direct crowding-out.** When subsidized/privileged SOEs expand within an industry, they can displace private competitors through market-share competition

and by absorbing scarce inputs (credit, land, procurement, or policy attention). If SOEs are not the most productive firms, this reallocation can reduce private-sector scale and weaken measured productivity in that industry.

- **Industrial SOE loans: downstream crowding-in.** When credit expands upstream production (often in input-supplying sectors), downstream firms can benefit from improved input availability, lower effective input costs, and higher reliability of supply. These gains are larger for more efficient downstream firms that can respond elastically by scaling production.
- **Infrastructure loans: broad crowding-in.** Infrastructure (transport, utilities, urban projects) raises the productivity of many firms simultaneously by reducing trade/commuting costs, improving market access, and relaxing local bottlenecks. This is why infrastructure credit shows positive and comparatively widespread effects on private-firm scale and efficiency.

7.4 Main implication (what to remember for research and policy)

The paper’s main lesson is that “government credit” is not a single treatment: **its welfare consequences depend on credit type and network position.** Evaluating development banking therefore requires: (i) separating *industrial SOE credit* from *infrastructure credit*, and (ii) tracing equilibrium effects through *production networks* rather than only examining directly treated firms.

7.5 Takeaway

CDB credit is double-edged: industrial lending to SOEs crowds out same-industry private firms but crowds in downstream private activity via supply-chain spillovers, while infrastructure lending broadly crowds in private-sector expansion and efficiency.